Machine Learning

By: Ahmad Masarrat
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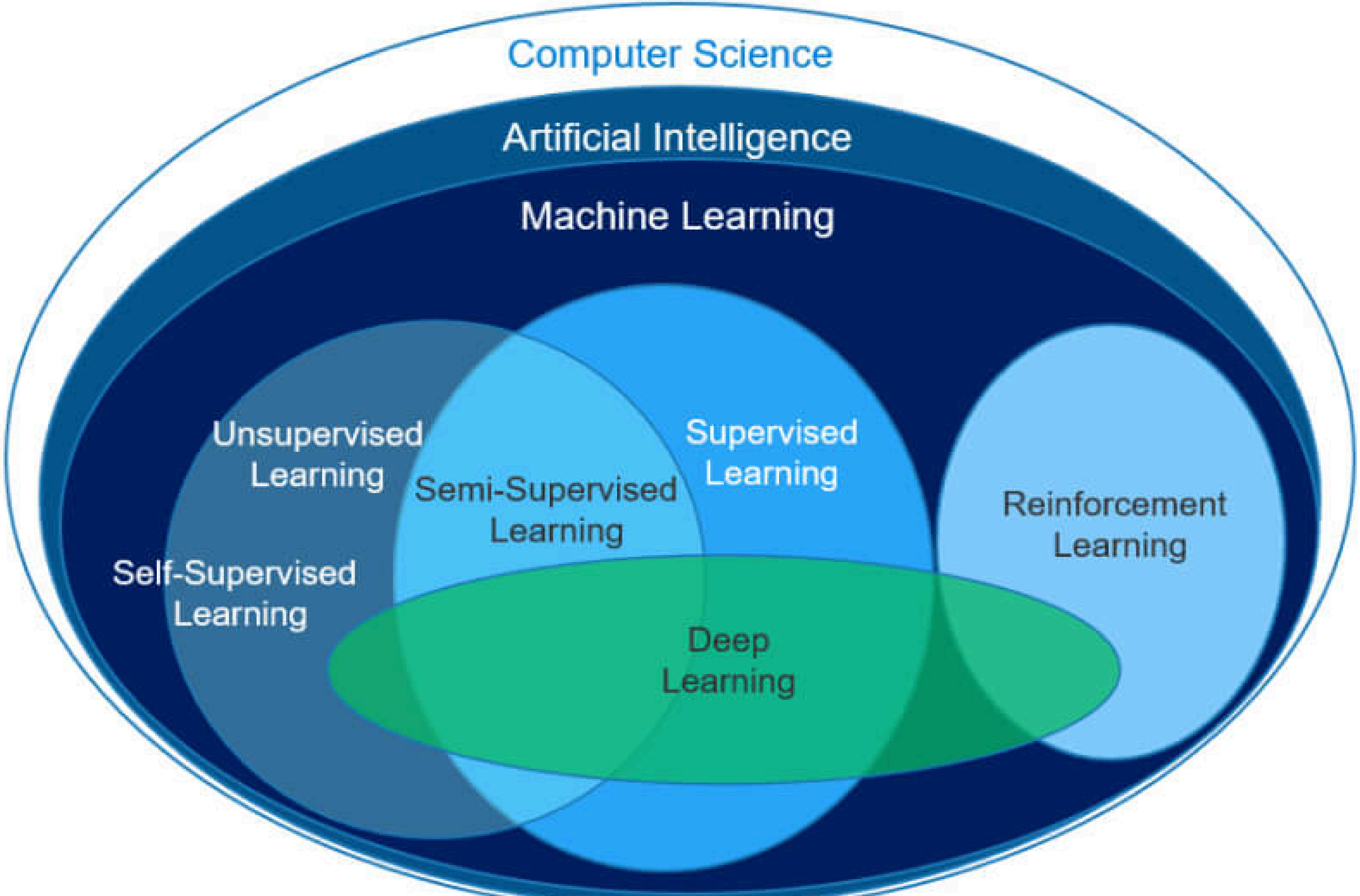
Outline:

Supervised Learning

Unsupervised Learning



by **Vlad Mysla** on Medium

Supervised Learning

Key Componets:

Features (X)
Inputs

Model
Mapping

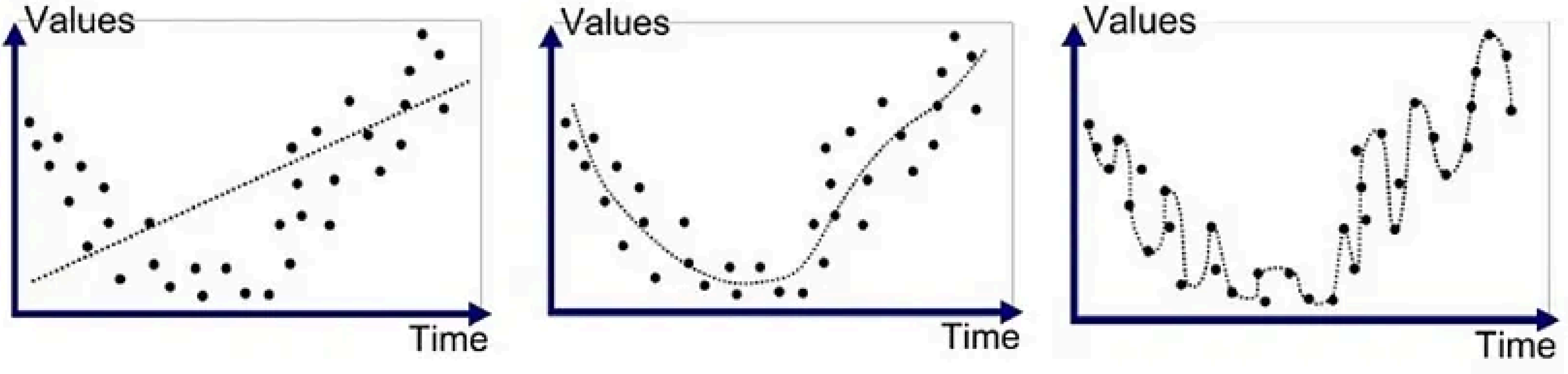
Target(y)
to predict

Types of Supervised Learning:

Classification
discrete

Regression
continues

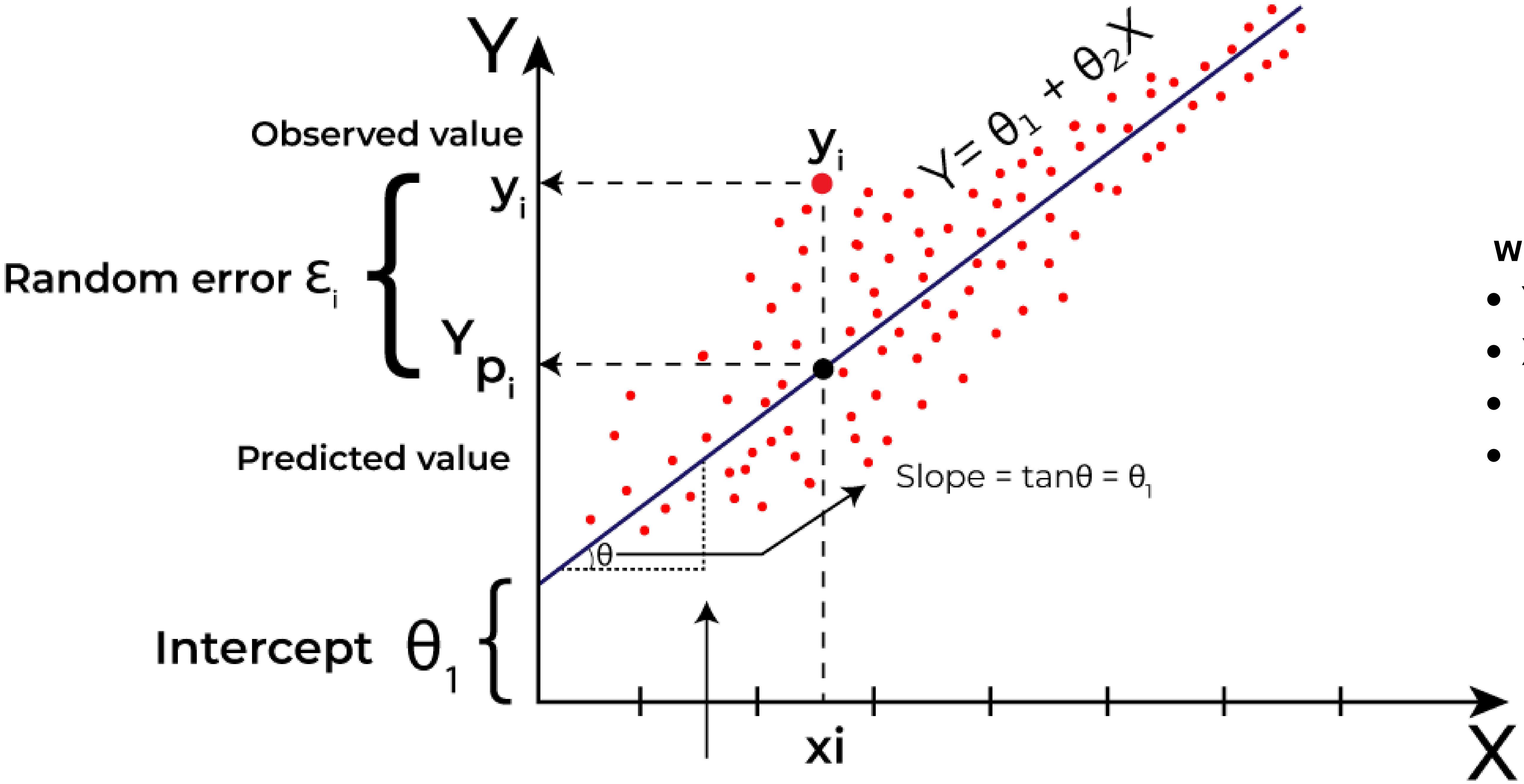
Underfitting vs Overfitting



Underfitted | Good Fit/Robust | Overfitted

Key Algorthims

Linear Regression



where:

- Y is the dependent variable
- X is the independent variable
- β0 is the intercept
- β1 is the slope

by:geeksforgeeks

Key Algorthims

Types Of Linear Regression

Simple
Linear regression

$y=\beta 0+\beta 1X$

Multiple
Linear regression

$y=\beta 0+\beta 1X1+\beta 2X2+.........\beta nXn$

Assumptions Of Linear Regression

Multiple LR

Simple LR

Linearity

Independence

Homoscedasticity

Normality

No multicollinearity

Additivity

Feature Selection

Cost Function for Linear Regression

Mean Squared Error (MSE)

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

n = Number of Data points

Yi = Observed Values

Y = Predicted Values

Evaluation of Linear Regression

Mean Squared Error (MSE)

$$\frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{\Sigma(y_i - \hat{y}_i)^2}{N - P}}$$

Coefficient of Determination (R-squared)

$$R^2 = 1 - \left(\frac{RSS}{TSS}\right)$$

Regularization For Linear Regression

Lasso Regression
(L1 Reg)

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(\widehat{y_i} - y_i) + \lambda\sum_{j=1}^{n}|\theta_j|$$

Ridge Regression
(L2 Reg)

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(\widehat{y_i} - y_i) + \lambda\sum_{j=1}^{n}\theta_j^2$$

Elastic Net Regression

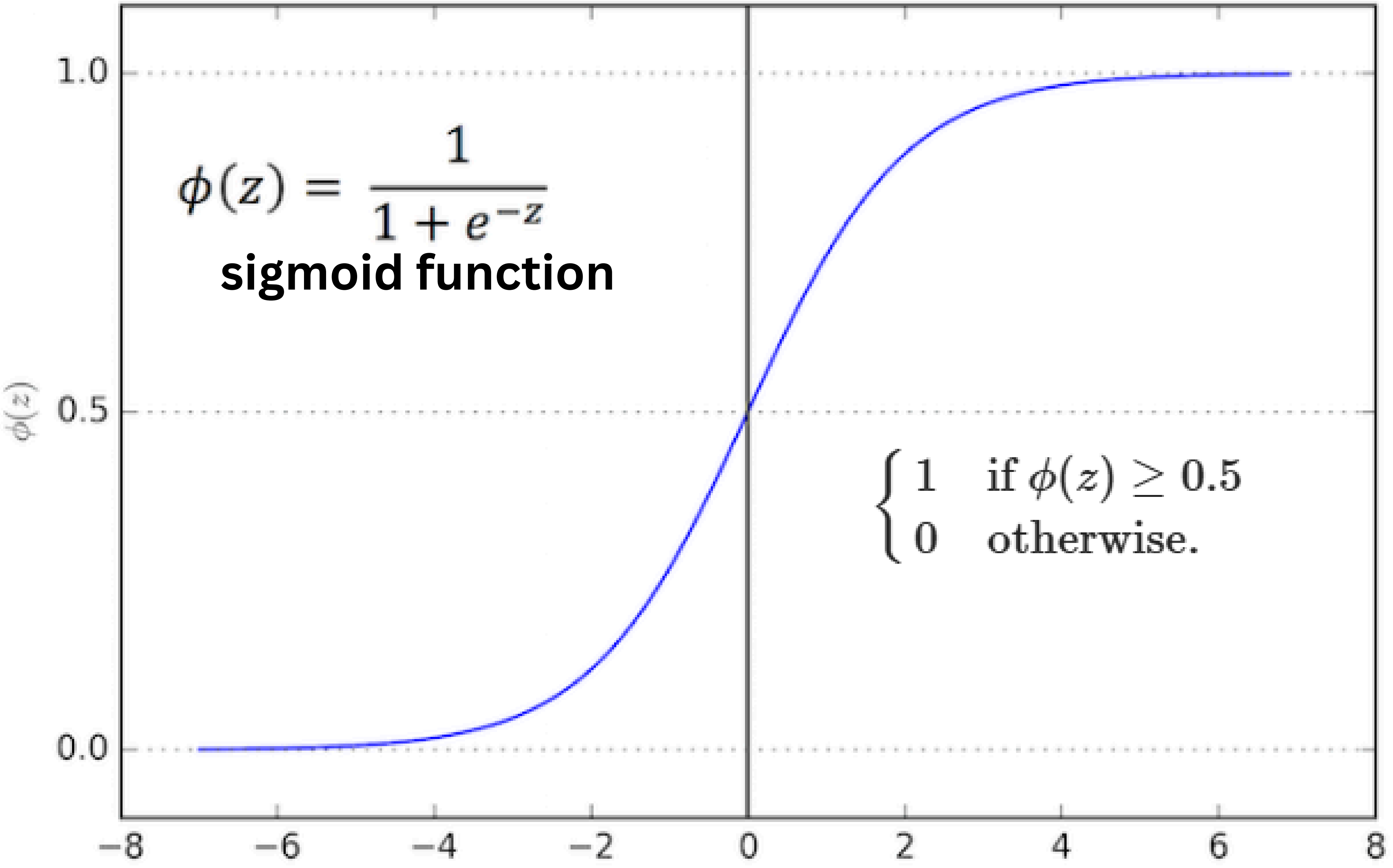
$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(\widehat{y_i} - y_i) + \alpha\lambda\sum_{j=1}^{n}|\theta_j| + \frac{1}{2}(1-\alpha)\lambda\sum_{j=1} n\theta_j^2$$

Key Algorthims

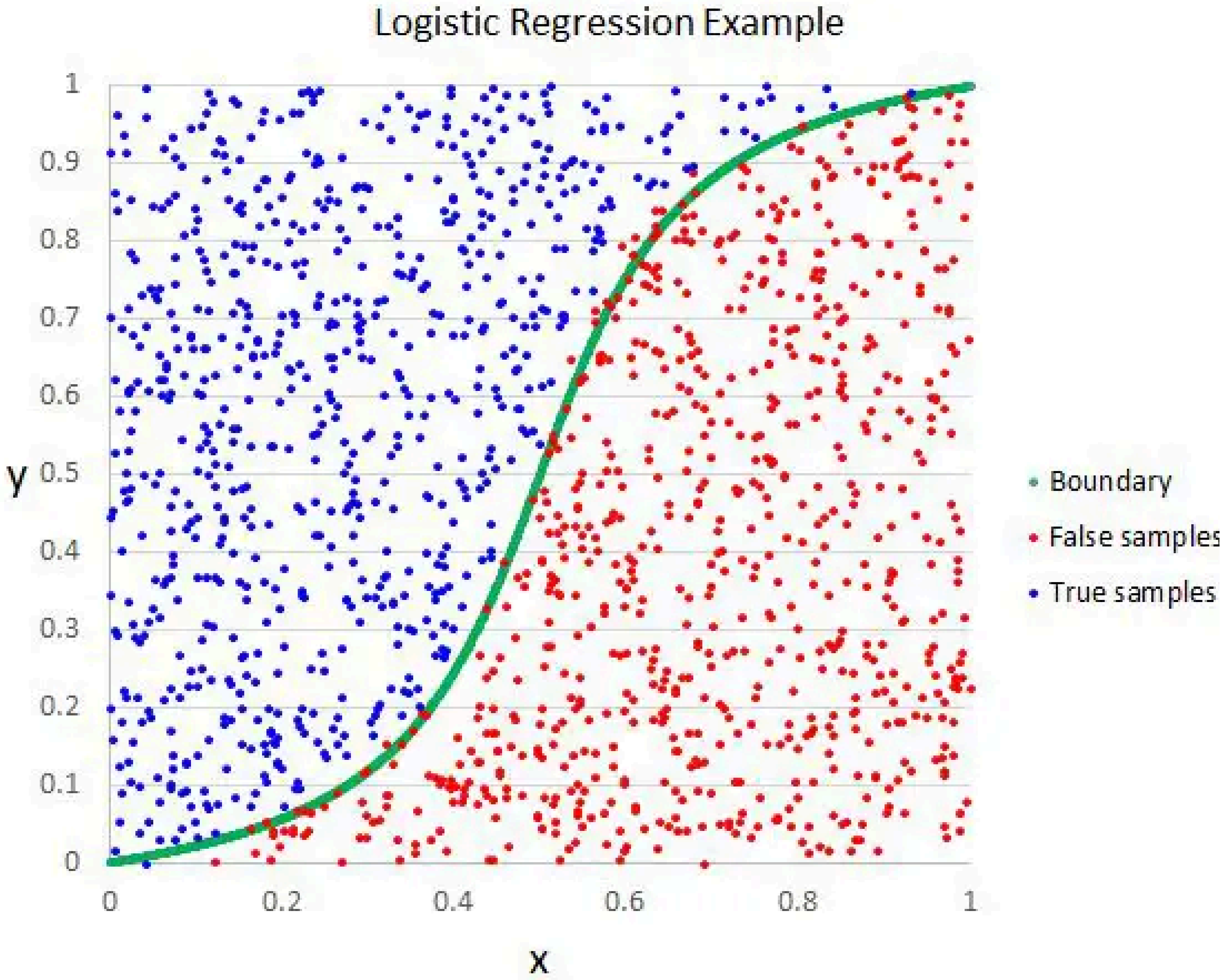
Logistic Regression

w·X+b

z : input



by:rasbt.github.io


Logistic Regression Example
y
x
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
0.2
0.4
0.6
0.8
1
Boundary
False samples
True samples

Assumptions Of
Logistic Regression

Independent observations
Binary dependent variables
No outliers
Large sample size

Evaluation

Confusion Matrix

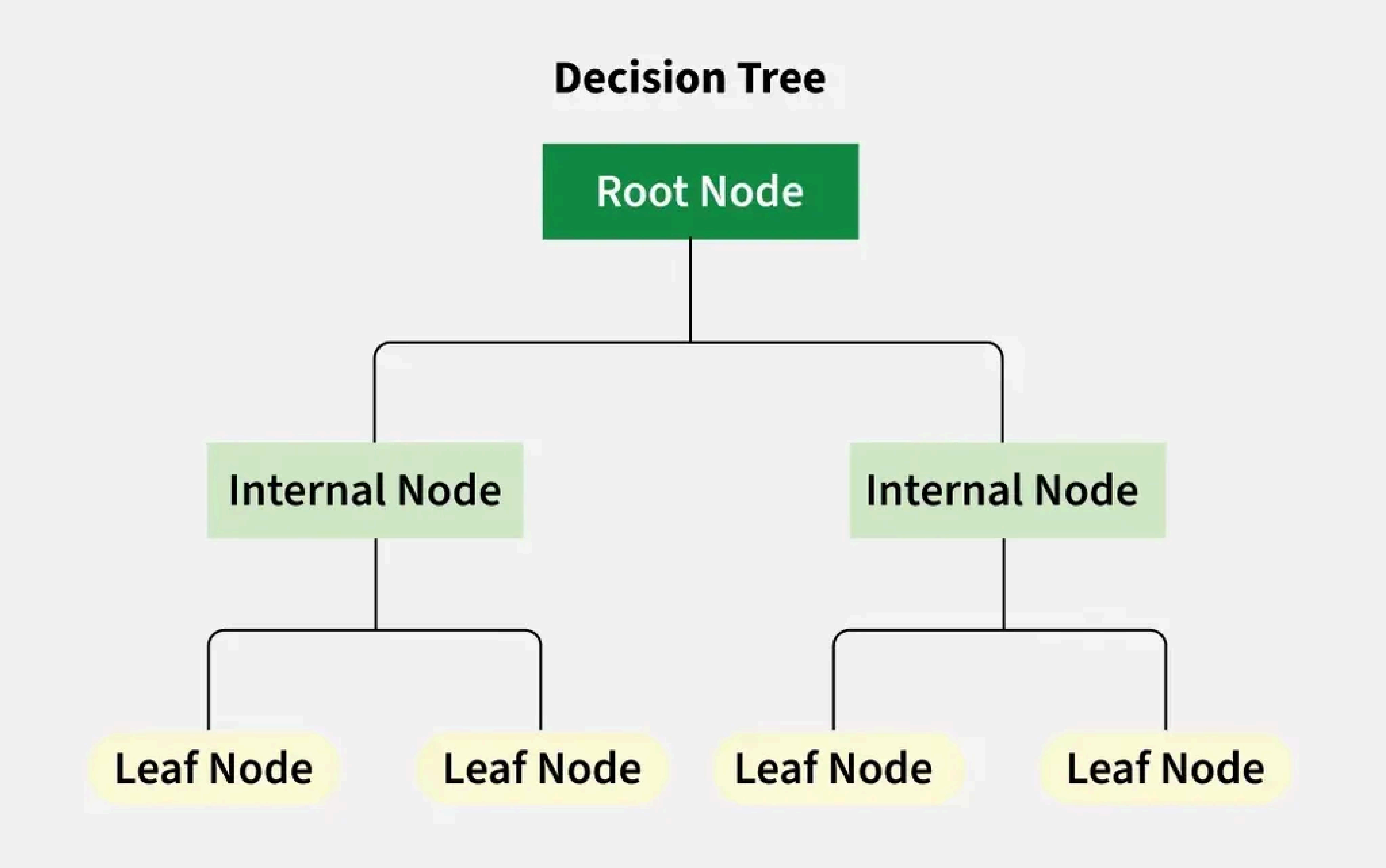
	1 (Predicted)	0 (Predicted)
1 (Actual)	True Positive	False Negative
0 (Actual)	False Positive	True Negative

$$preision = \frac{TP}{TP + FP}$$

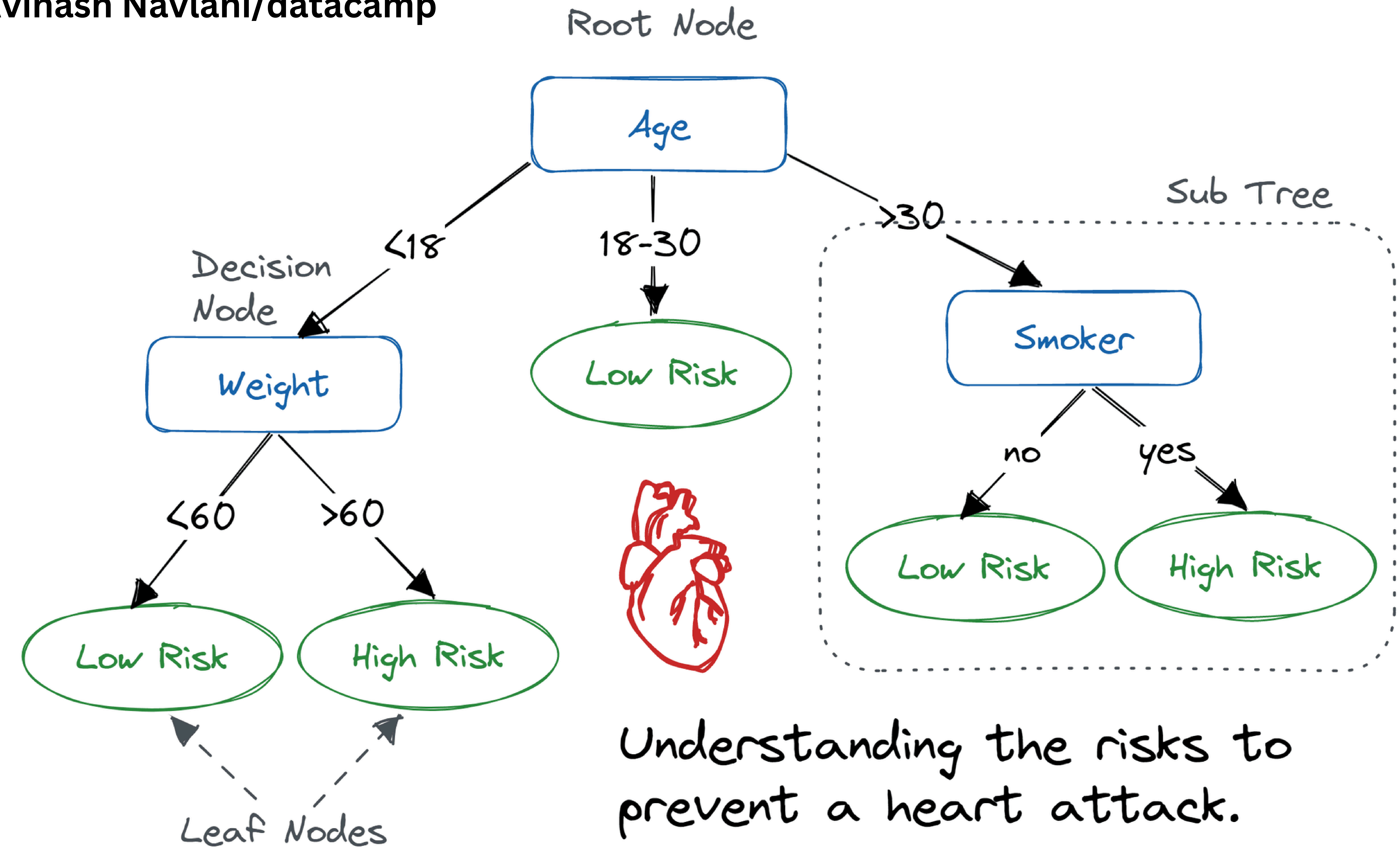
$$recall = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \times precision \times recall}{precision + recall}$$

Decision Trees



by:Avinash Navlani/datacamp



But How?

Information Gain: Uses Entropy to make decisions

Entropy: the uncertainty/randomness in the data.

- Entropy quantifies impurity in a dataset.
- Information Gain measures how much entropy is reduced after a split.
- Higher Information Gain → better feature for splitting in decision trees.
- Decision Trees use Information Gain (or Gini Index) to determine the best feature for classification.

Types of Decision Trees

Classification Trees
target variable is discrete

Regression Trees
target variable is continuous

Advantages & Disadvantages of Decision Trees

Adv

- Simplicity and Interpretability
- Versatility
- No Need for Feature Scaling
- Handles Non-linear Relationships

disAdv

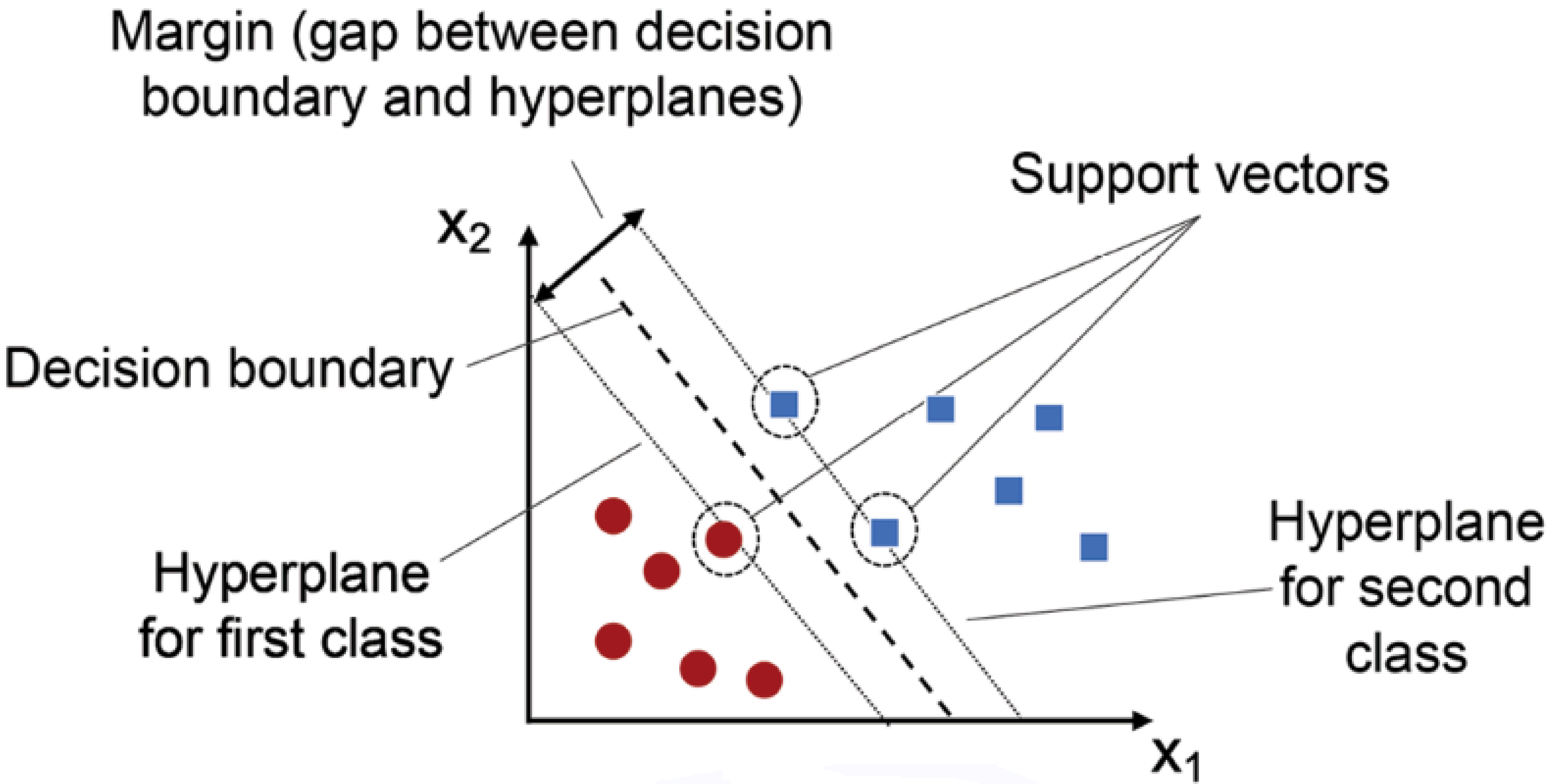
- Overfitting
- Instability
- Bias towards Features with More Levels

Key Algorthims

SVM

Support Vector Machine

$$w^T x + b = 0$$



Mythology

Hyperplane

Margin

Soft Margin

Hinge Loss

Kernel

C

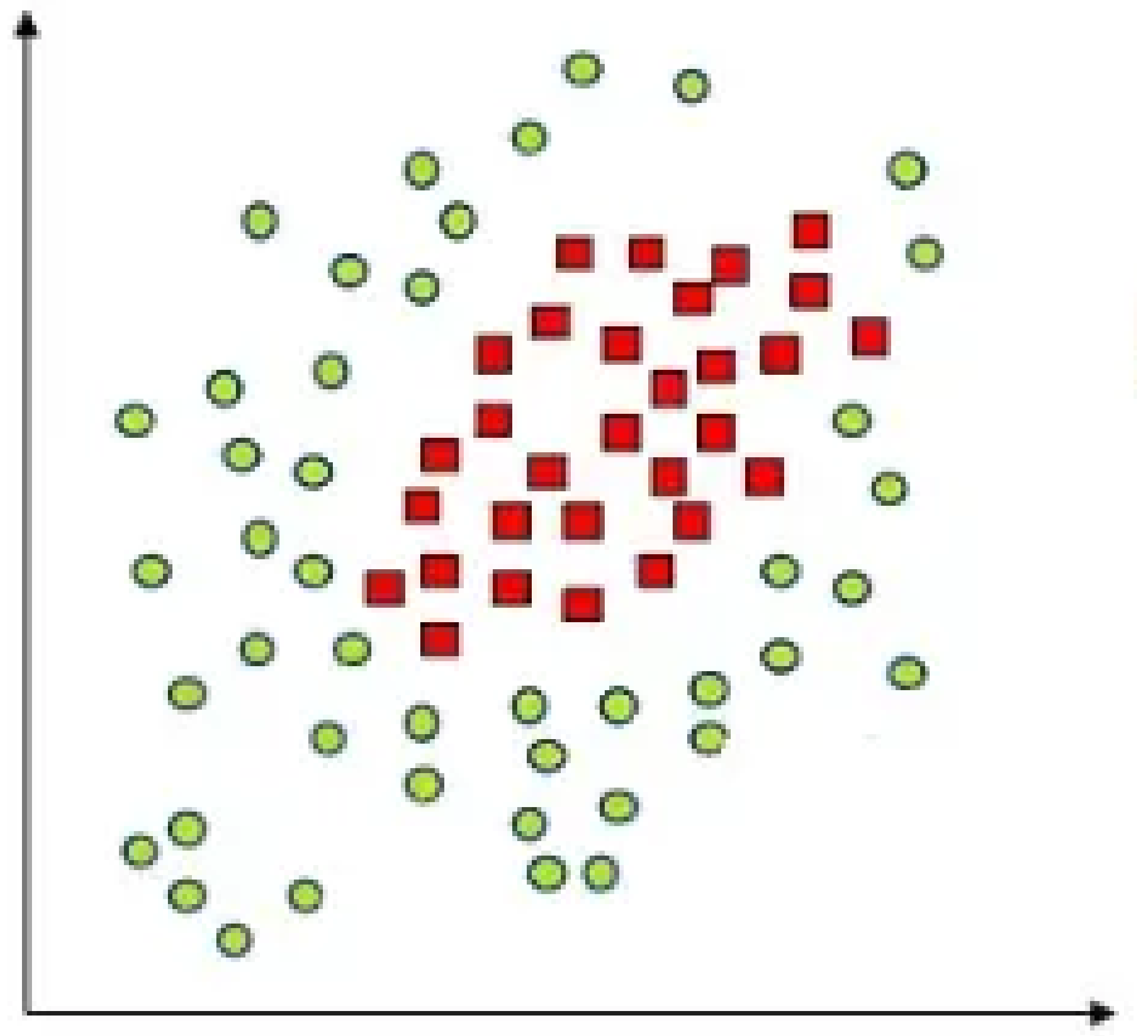
ζ

Support Vectors

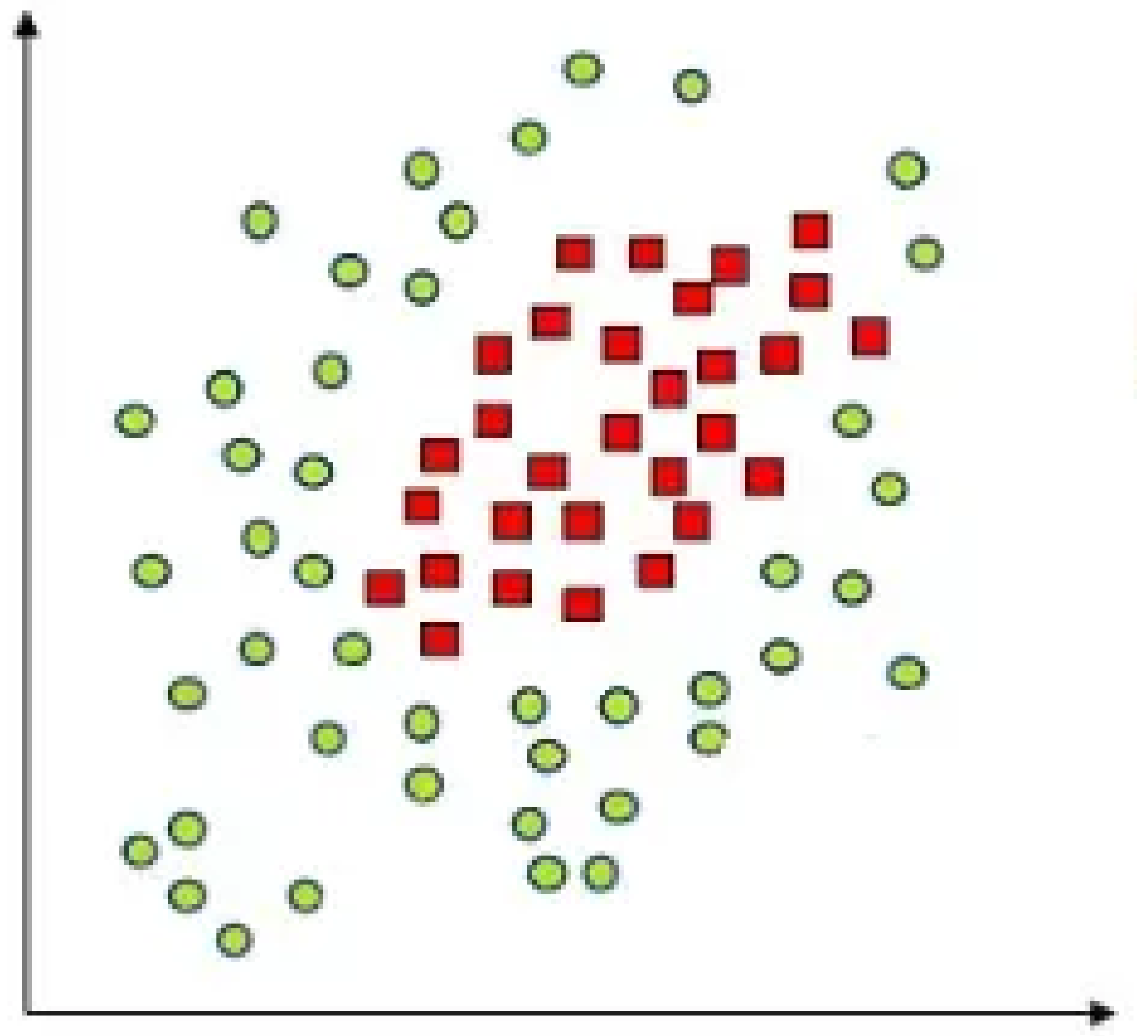
Hard Margin

Dual Problem

what if the data is not linearly separable?



what if the data is not linearly separable?

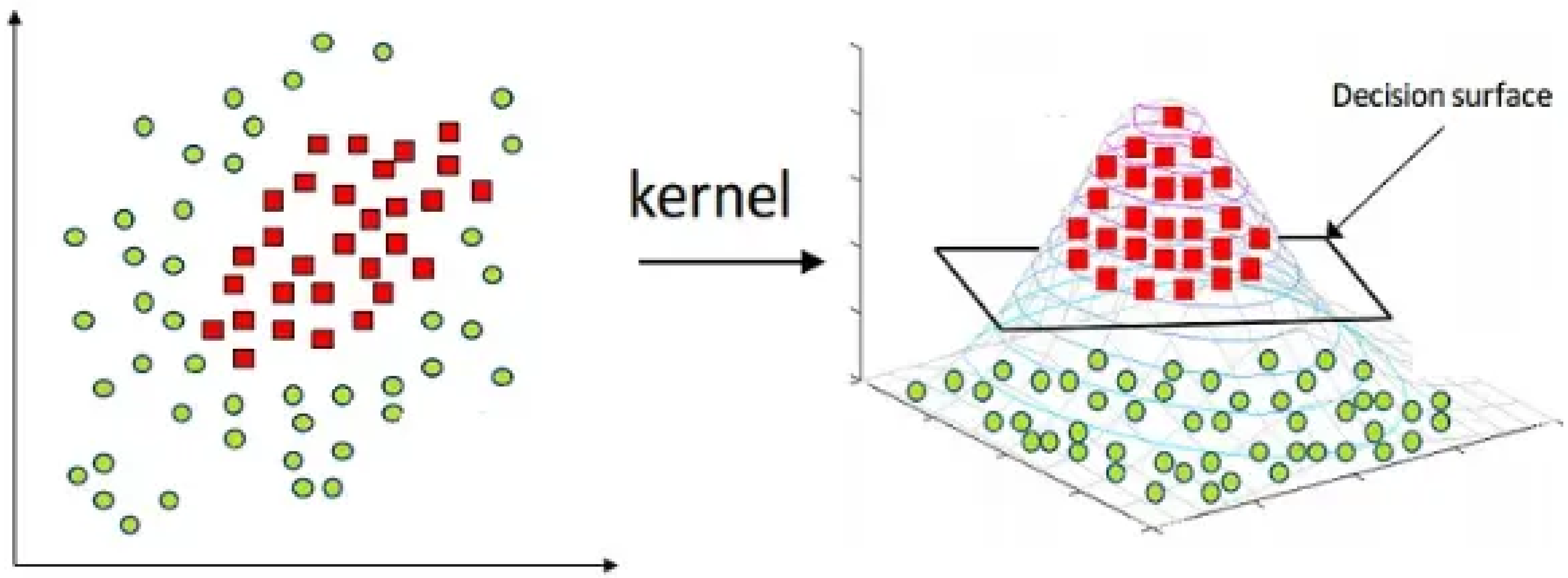


Linear Kernel

Polynomial Kernel

Radial Basis Function (RBF) Kernel

what if the data is not linearly separable?



Common Mistakes & Best Practices

mistakes	best practices
Data Leakage	Start Simple
Ignoring Class Imbalance	Feature Importance
Overfitting	Pipeline Everything

Unsupervised Learning

it knows nothing

Unsupervised Learning

Key Components

Features (X)

Target Variable?

Unsupervised Learning

Key Components

Features (X)

Target Variable?
there isn't.

Unsupervised Learning

Types

Clustering

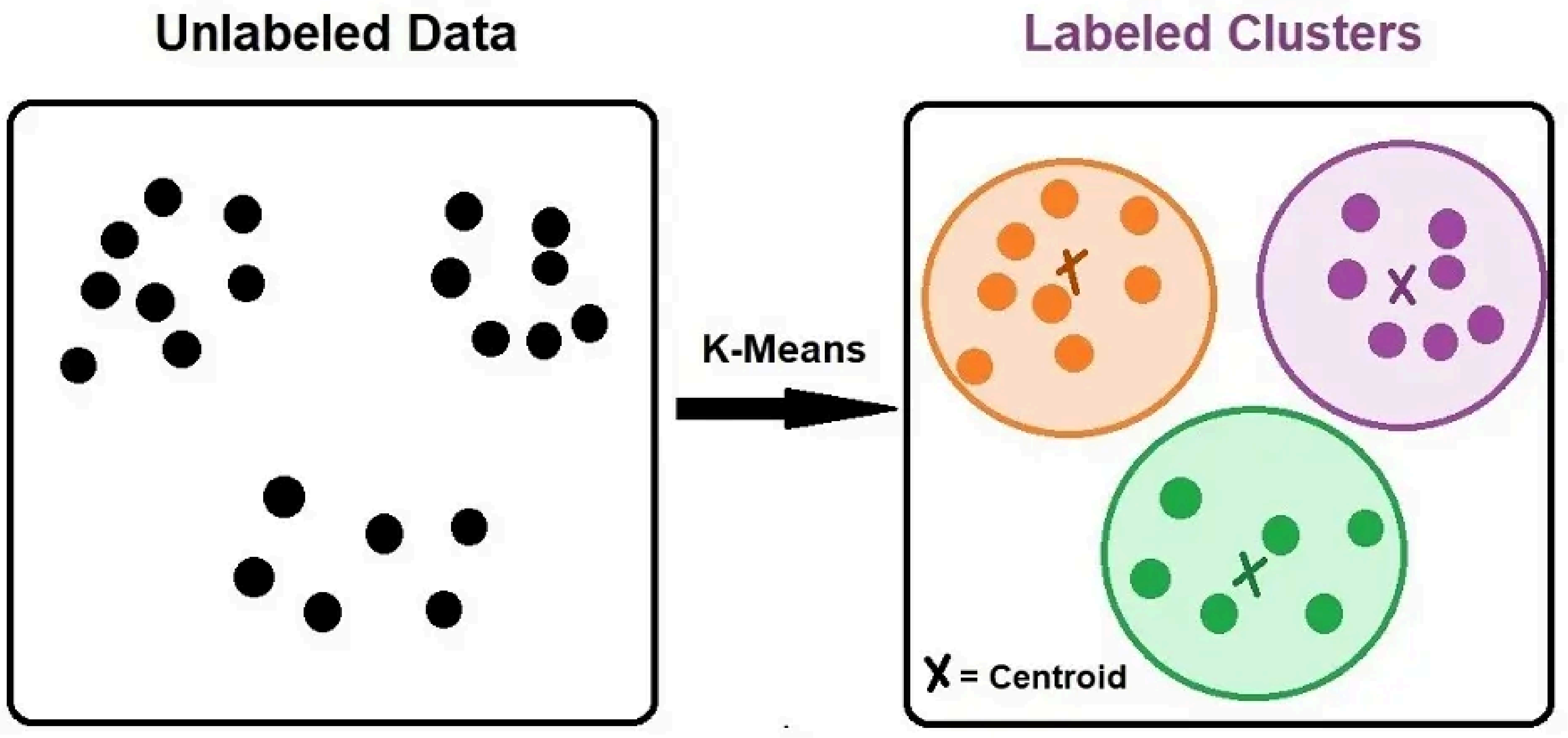
Dimensionality Reduction

Anomaly Detection

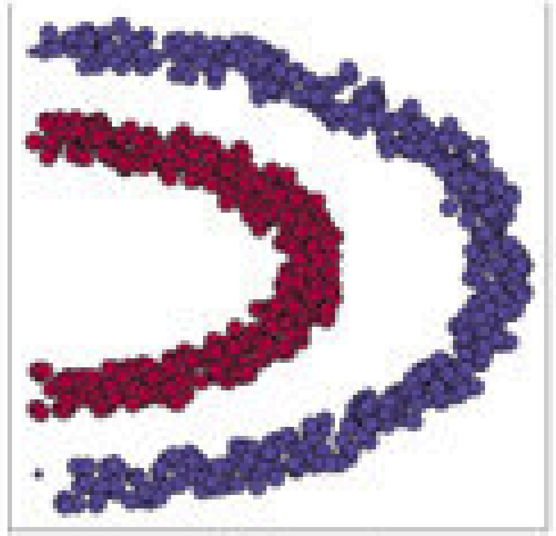
Association Rule Learning

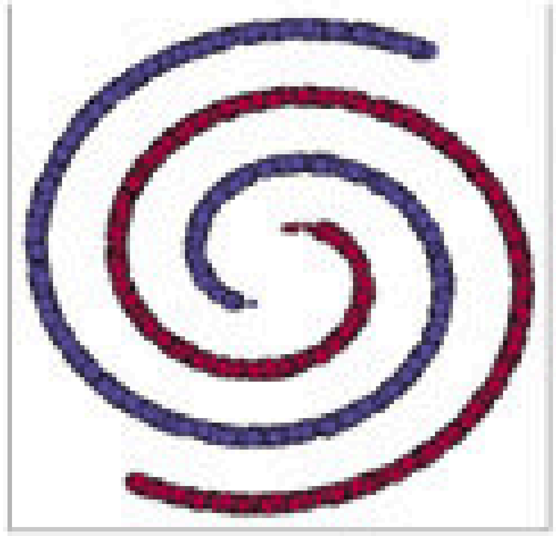
Density Estimation

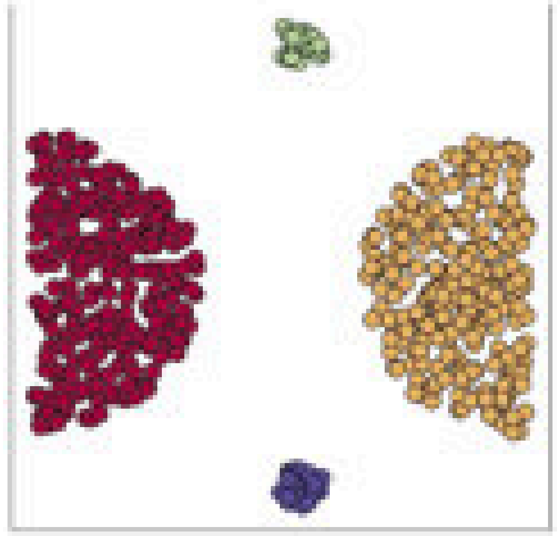
Clustering

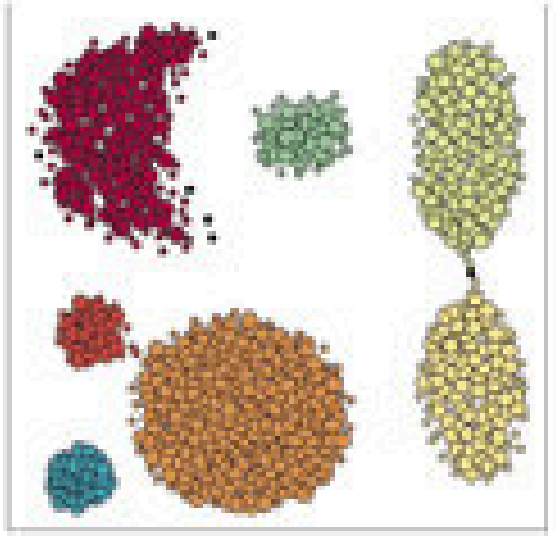


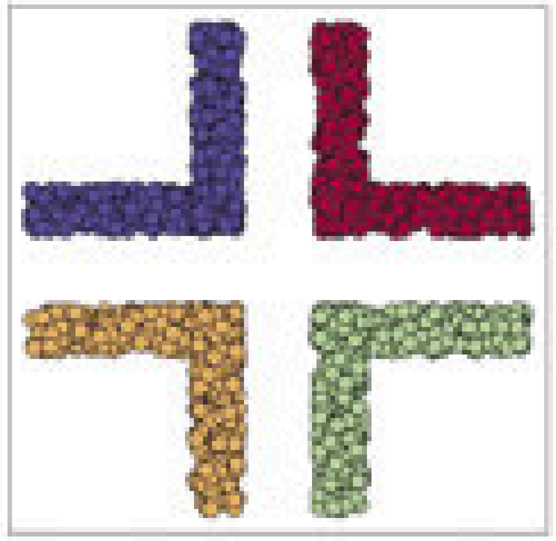
by:<u>Nisha Arya</u> on EJable

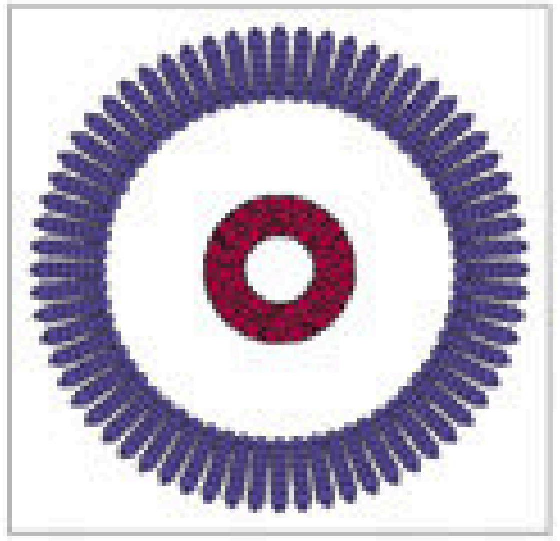

Half-Kernel

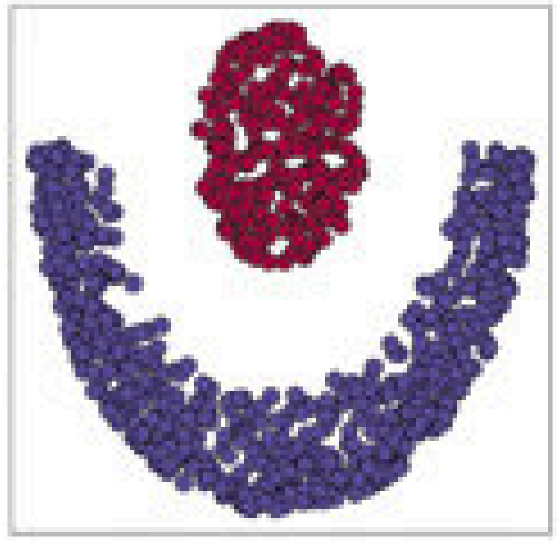

Two-Spirals

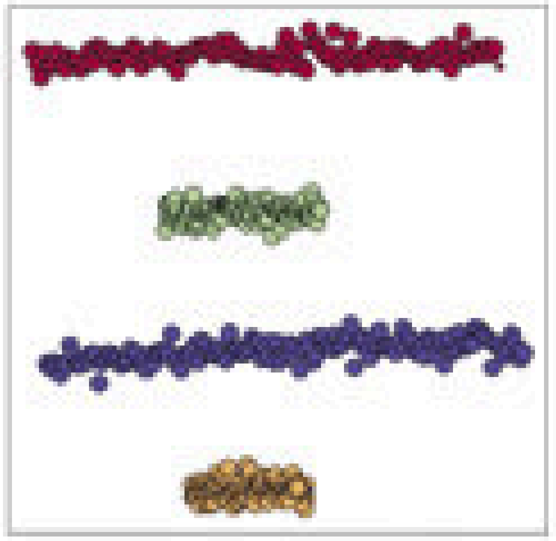

Outliers


Aggregation

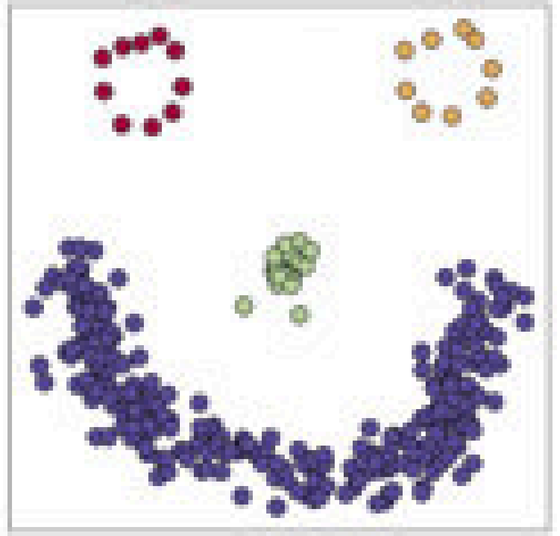

Corners

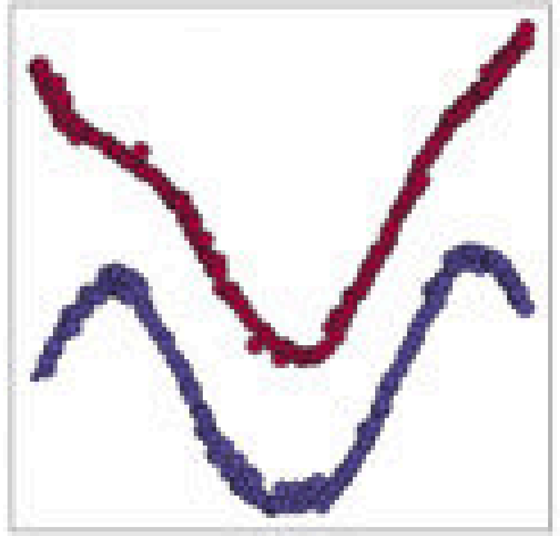

Cluster-in-cluster

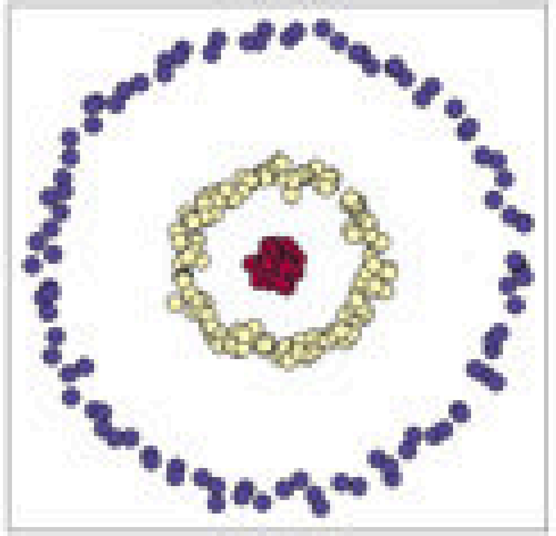

Crescent ful moon


Zelnik5


Moon


Face


Wave


Zelnik1

- **by: Ali Şenol/researchgate**

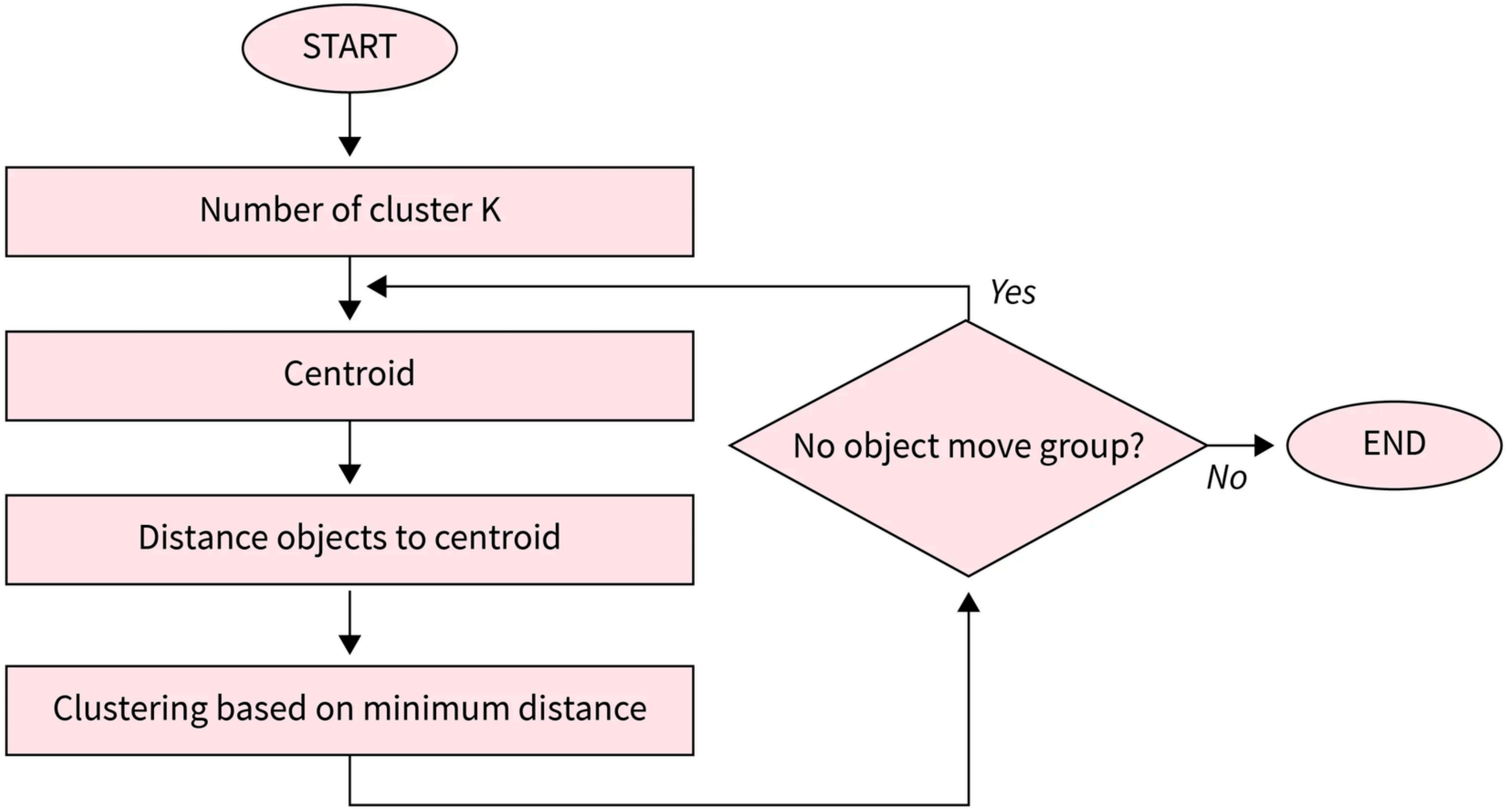
Clustering Types

Hard Clustering
Categories

Soft Clustering
Probabilities

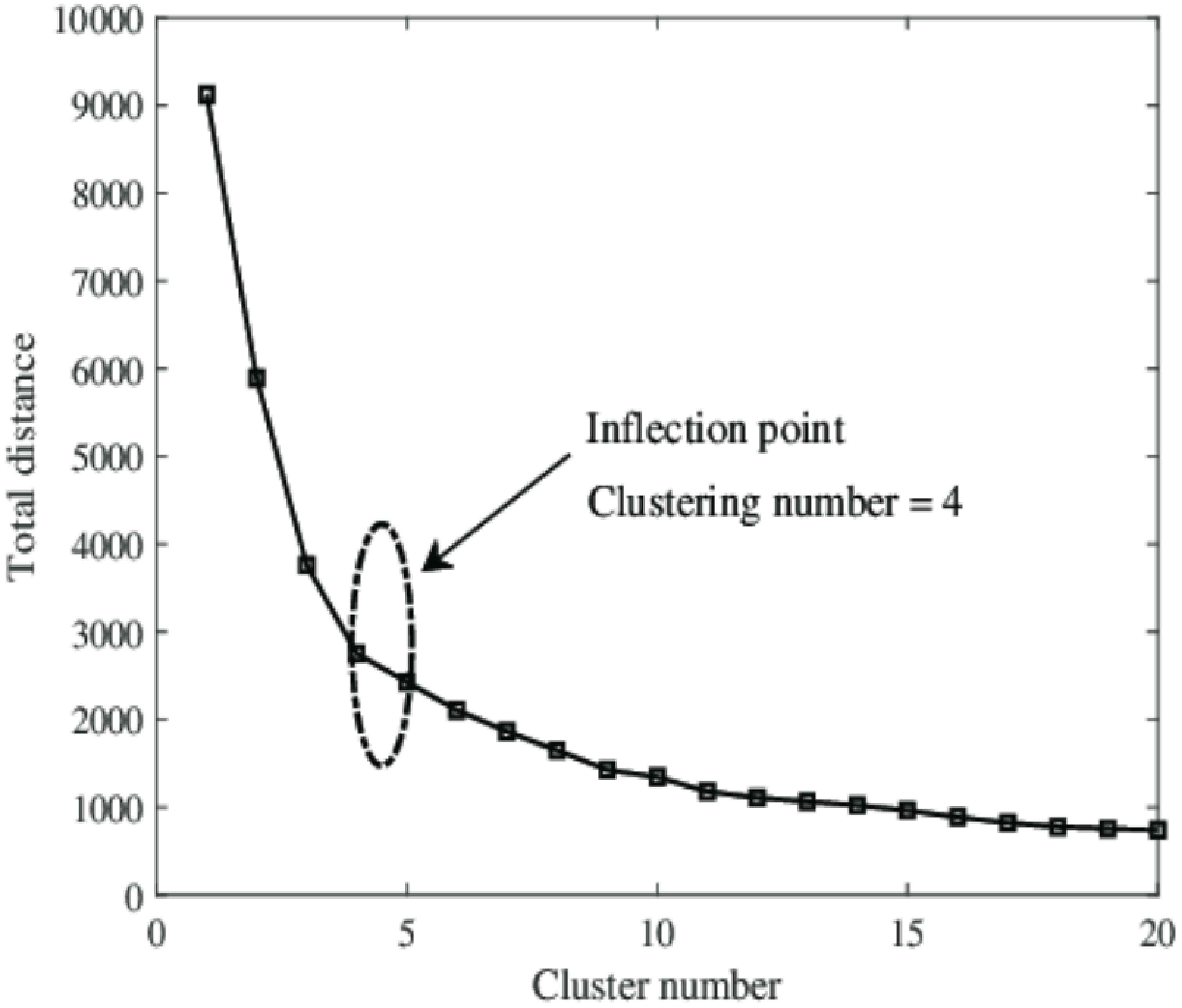
TYPES OF CLUSTERING ALGORITHMS

Centroid-based Clustering (Partitioning methods)



but what's K?
use your brain or elbow

elbow?



yes

[research gate]

Density-based Clustering (Model-based methods)

- Can find arbitrarily shaped clusters
- Handles noise and outliers well
- Excels with clusters of different sizes and shapes
- Ideal for datasets with irregularly shaped or overlapping clusters
- Effectively manages both dense and sparse data regions
- Focus on local density allows detection of various cluster morphologies

DBSCAN

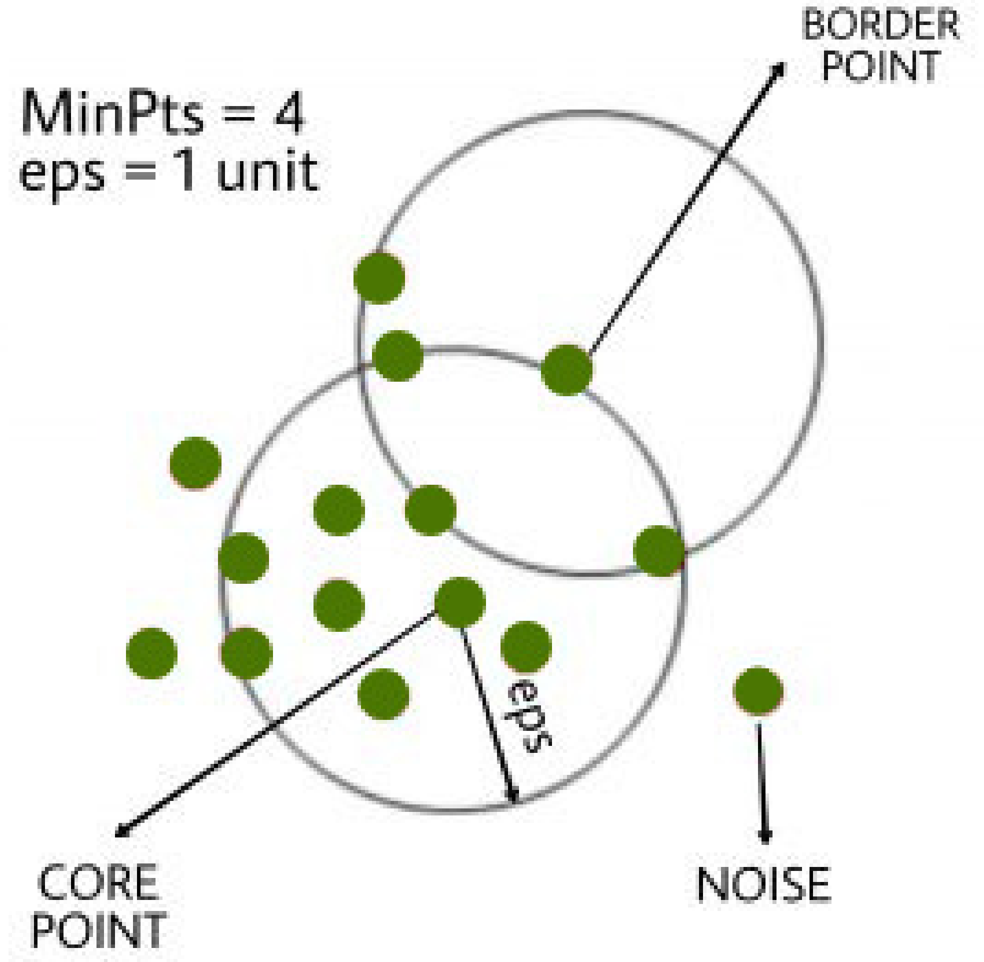
density-based clustering algorithm

key parameters

eps
MinPts

Steps in DBSCAN

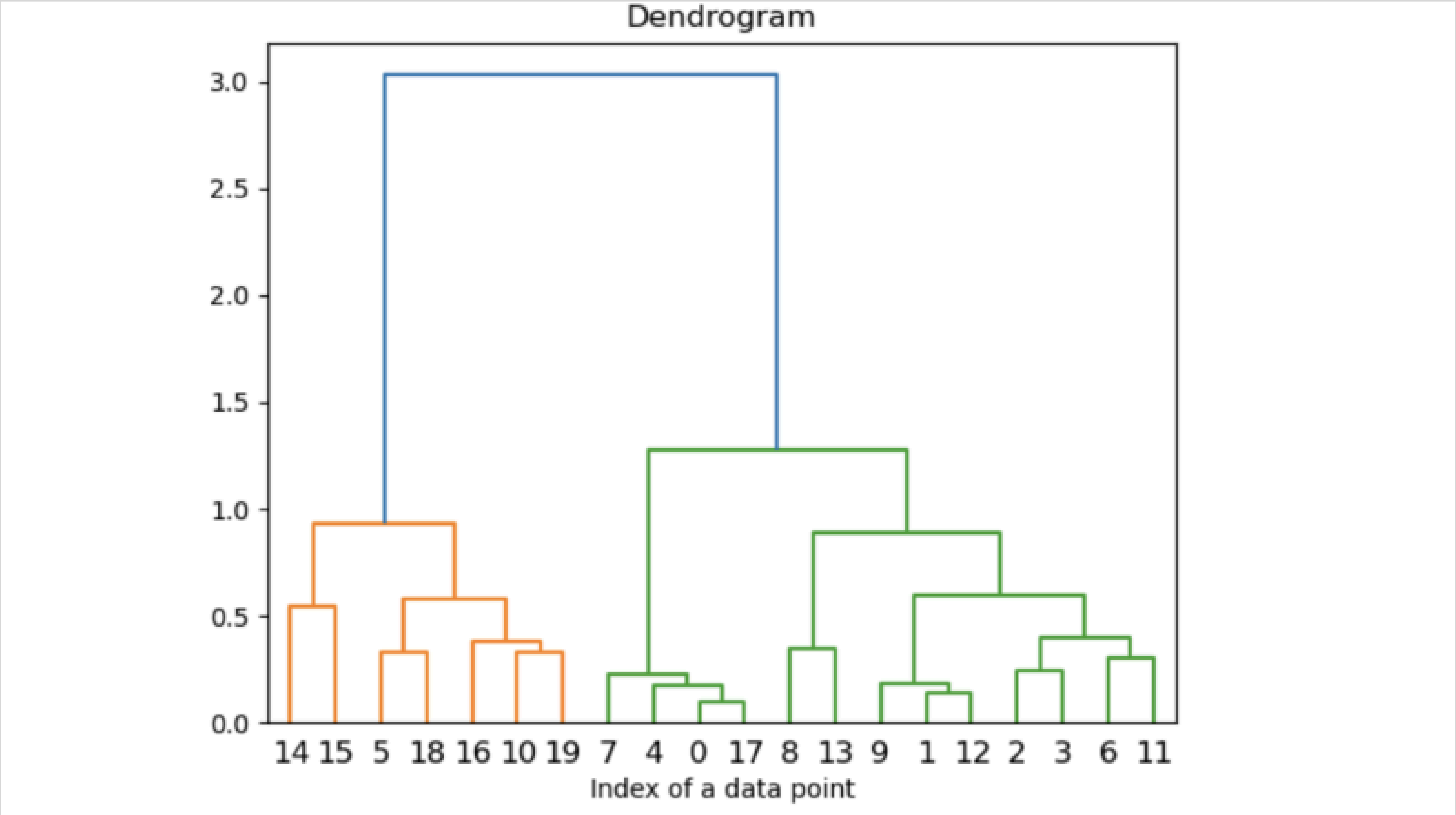
Identify Core Points
Form Clusters
Density Connectivity
Label Noise Points



Connectivity-based Clustering (Hierarchical clustering)

Divisive Clustering

Agglomerative Clustering



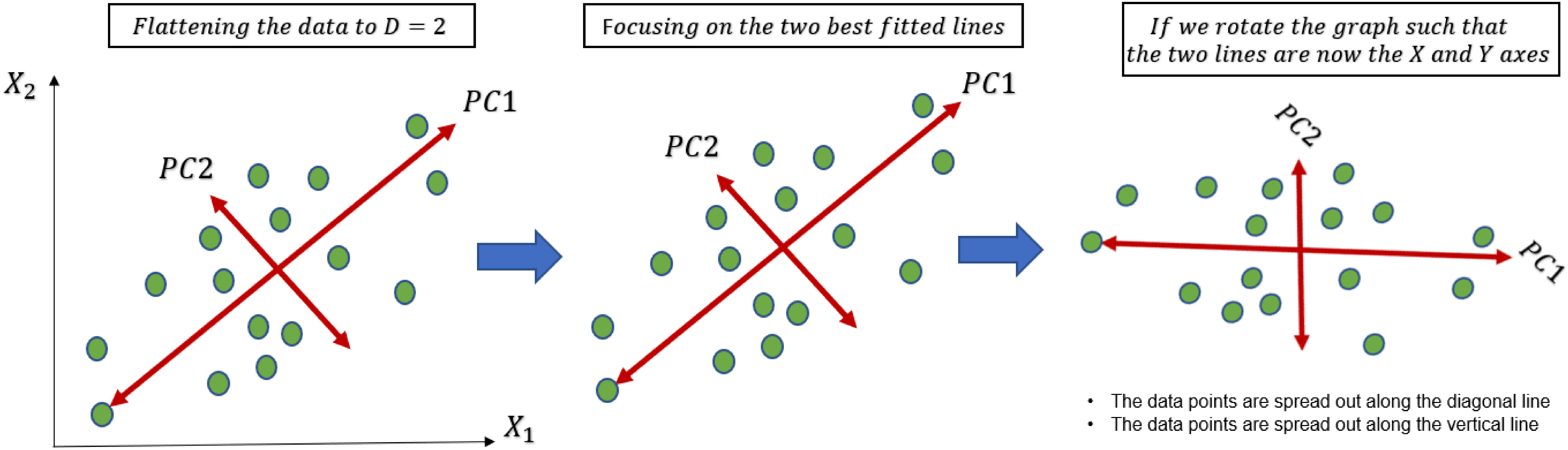
Soft Clustering

most popular algorithm is Gaussian Mixture Model.

Fuzzy Clustering
to belong to multiple clusters with varying degrees of membership.

Dimensionality Reduction using PCA

(Principal Component Analysis)



Michael W

Dimensionality Reduction using

PCA

(Principal Component Analysis)

Steps:

Step 1: Standardize the Data
mean=0, std=1

$$Z = \frac{X-\mu}{\sigma}$$

μ is the mean of independent features
σ is the standard deviation

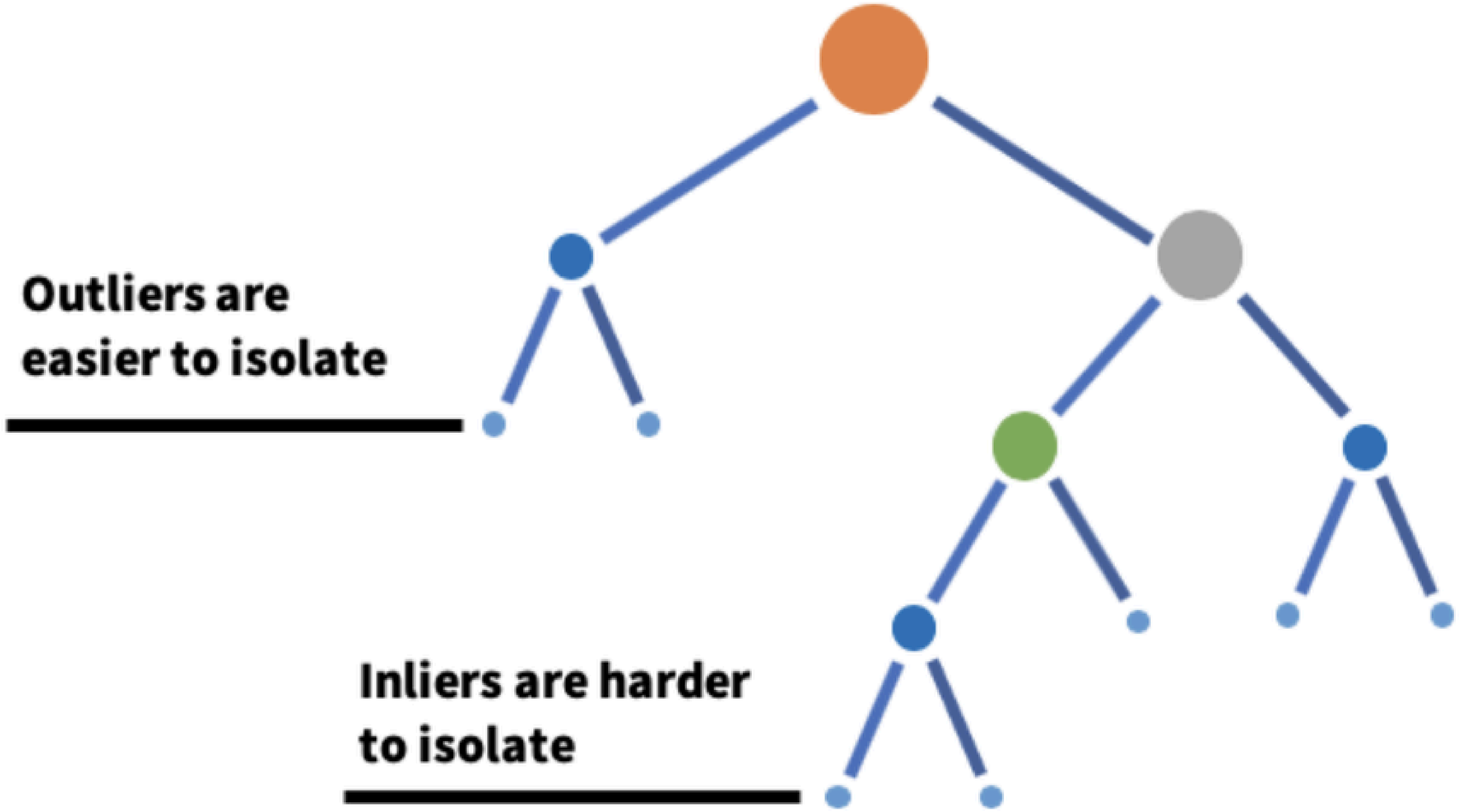
Step 2: Find Relationships
use Covariance

Step 3: Find Principal Components, choose top 3

PC1

PC2

Anomoly Detection using **Islation Forest**



Key Concepts

-Isolation: focus on uniqueness

-Partitioning: randomly selecting features and splits

-Anomaly Score: how many splits were needed to isolate it?

by:James Verbus/linkedin

Anomoly Detection
using **Islation Forest**

applications

Detecting Malicious Network Traffic

Identifying Fraudulent Transactions

Detecting Anomalous Patient Data

Common Mistakes & Best Practices

Mistakes

Ignoring Feature Scaling

Forcing Wrong k in K-Means

Best Practices

Validate Clusters

Layer Techniques

Visualize Anomalies

THANK YOU